

Highlights

- Free-coordinate registration jointly estimates pose, warp, and intrinsics.
- Observability-guided warp acts only in the provably observable subspace.
- Matches pose champions on BOP yet uniquely survives endoscopy (4.23 mm).
- Learned front-end + ICP reaches ADD-S 0.987 at five-fold speed.
- SOTA collapses out-of-distribution in endoscopy; FreeReg does not.